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Pyrolysis of municipal sewage sludge for bioenergy production: Thermo-kinetic studies, evolved gas analysis, and techno-socio-economic assessment

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ABSTRACT

In this study, a comprehensive investigation was conducted to use municipal sewage sludge (MSS) as a promising feedstock for bioenergy production via pyrolysis process. Using thermogravimetric analysis (TGA), MSS was subjected to thermal decomposition experiments at four different heating rates of 5, 10, 30 and 50 °C/min. TGA curves were divided into three distinctive stages, namely drying zone ($T \le 200$ °C), active pyrolysis zone ($200 < T < 600$ °C), and char decomposition zone ($T \ge 600$ °C). Moreover, the data were used to analyze thermo-kinetic parameters through Flynn-Wall-Ozawa (FWO), Kissenger-Akahira-Sunose (KAS), and Starink methods, where the average values of E_a (126.62–136.92 kJ/mol), Gibbs free energy (159.19–159.61 kJ/mol), and calculated high heating value (HHV, 16.47 ± 0.03 MJ/kg) showed the considerable bioenergy potential of the low-cost biomass. The low difference between E_a and ΔH ($\sim$ 5 kJ/mol) showed that product formation was favorable during pyrolysis. In addition, for the first time, the comparative study of the results predicted from support vector regression (SVR) model and the experimental data, showed a satisfactory agreement ($R^2 > 0.9999$) and accurate results regarding different train-test data categories. Further employed was Pyrolysis–Gas Chromatography/Mass Spectroscopy (Py-GC/MS) at 700 °C to characterize the potential chemical products, which indicated the presence of a range of aromatic and aliphatic hydrocarbons, nitrogen-containing compounds, alcohol, furans and sulfur compounds. Finally, three alternative scenarios associated with the planned project were presented and techno-socio-economic assessment (TSEA) of the alternative scenarios were evaluated, for the first time, by means of economic indexes and considering social aspects. Net present value (NPV) of all alternative scenarios of the pyrolysis plant over the 20-year plant lifetime was positive compared to the base case, indicating that the project was feasible. Sensitivity analysis of the optimistic scenario showed that the profitability of pyrolysis plant was highly dependent on bio-oil selling price and total production cost.

1. Introduction

Over the last two decades, there has been increasing interest toward renewable energy sources such as wind, solar energy, geothermal energy, fuel cells, hydrothermal energy and biomass. The exploitation of renewable energy and improvement of energy efficiency are, two approaches to addressing the rising demand for energy in different industries [1]. Biomass, the organic-rich material, is expected to contribute to future sustainable development of energy systems and has the potential to be converted into bio-oil and bio-char through applying suitable conversion technologies [2,3]. Municipal sewage sludge (MSS), the residue produced by the wastewater treatment plants (WWTPs), is often considered as biomass. MSS production (dry weight) is on an increasing growth curve. For instance, in European Union, the production was 11.5 Mt (Million tons) in 2010, expected to rise to 13.0 Mt in 2020 [4]. Therefore, the management of MSS production in an environmentally-friendly manner is a major challenge around the world [5].

By means of thermochemical or biochemical conversion processes, MSS can be converted into energy source [2]. The biochemical conversion processes, such as anaerobic digestion, are time-consuming, and use enzymes or microorganisms to decompose biomass into lower molecular weight compounds such as methane-rich biogas [6]. The thermochemical conversion processes, including pyrolysis, combustion, gasification, and liquefaction, employ heat to produce energy products

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Nomenclature (symbols/abbreviations)

$G(\alpha)$	Conversion rate equation
α	Conversion degree (%)
β	Heating rate (°C/min)
K_B	Boltzmann constant (1.381×10^{-23} J/K)
h	Planks constant (6.626×10^{-34} Js)
A	Pre-exponential factor (min^{-1})
E_a	Activation energy of pyrolysis reaction (kJ/mol)
R	Universal gas constant (0.008314 kJ/mol.K)
T	Absolute temperature (K)
T_m	DTG peak temperature (K)
ΔH	Enthalpy change (kJ/mol)
ΔG	Gibb's free energy (kJ/mol)
ΔS	Entropy change (J/mol·k)
k	Reaction rate constant
m_o	Initial sample weight in TGA (wt %)
m_t	Sample weight at time t in TGA (wt %)
m_r	Residual mass after TGA (wt %)
VM	Volatile matter (wt %)
FC	Fixed carbon (wt %)
HHV	High heating value (MJ/kg)
SVR	Support vector regression
λ_i	Experimental value
β_i	Predicted values
n	Number of data points
MSE	Mean square error
TSEA	Techno-socio-economic assessment
NPV	Net present value
ROI	Return on investment
IRR	Internal rate of return

from MSS [7,8]. The latter process is superior to the former owing to its higher conversion efficiency [9–11]. Among thermochemical processes, pyrolysis has been increasing attention as an acceptable route for waste disposal due to product selectivity, process scalability and versatility. Pyrolysis, the thermal degradation process of material in the absence of oxygen, results in a decomposition solely determined by chemical characteristics of the sample, using an endothermic reaction, yielding bio-gas, bio-oil and bio-char. Recently, the interest towards pyrolysis of MSS, as a remarkable feedstock for bioenergy production, has shown rapid growth in literature [1,12,13]. The performance of this process highly depends on the sewage sludge (SS) composition which is mainly affected by the loading capacity of WWTPs, treatment method, seasonal variation, and the type of industrial plants connected to WWTPs [12]. Proximate, elemental and high heating value (HHV) analysis play important roles in evaluating the thermochemical conversion behavior of biomass. The volatile matter (VM), ash content, and fixed carbon (FC) of MSS (on dry bases) have a variance of approximately 36–75%, 15–61%, and average 8%, respectively [14], rendering it preferable for bioenergy production. The elemental analysis of MSS ranges between 45 and 55% C, 25–40% O, 5–12% N, 6–10% H, and 0.5–1.5% S [15–17]. The calculated HHV of SS was reported to range from 10.4 MJ/kg to 19.5 MJ/kg [18,19]. A few articles have been published on the thermo-kinetic behavior of SS pyrolysis [19–21]. Flynn-Wall-Ozawa (FWO), Kissenger-Akahira-Sunose (KAS), distribution activation energy, Coats-Redfern, and Starink are some well-known methods for determining the thermo-kinetic parameters of MSS which are highly dependent on the feedstock characterization, reaction mechanism, conversion point, and the selected kinetic method [10,20,22,23]. Naqvi et al. [20] studied FWO and KAS methods, and Huang et al. [23] studied distribution activation energy method for the pyrolysis of SS. Their studies showed the significant complexity of the biomass for pyrolysis process. The range of activation energy (E_a), enthalpy change (ΔH), and Gibbs free energy (ΔG) of MSS pyrolysis were 45.6–231.7 kJ/mol, 41.5–227.6 kJ/mol, and 53.1–295.2 kJ/mol in FWO method, and 41.1–231.7 kJ/mol, 37–227.6 kJ/mol, and 92.4–295.6 kJ/mol in KAS method at different conversion points, respectively [20].

To provide more information concerning pyrolysis products and the energy involvement of the sample, thermogravimetric analysis (TGA) coupled with evolved gas analysis technique was further investigated [24]. The Pyrolysis–Gas Chromatography/Mass Spectroscopy (Py-GC/MS) analysis of SS showed two major pyrolysis products (~60% of total products), namely benzene and its derivatives and $C7_p$ (the products whose carbon atoms exceeded 7). The TG-FTIR analysis also indicated that aromatic compounds were major gaseous products in the pyrolysis of SS [24].

Pokorna et al. [15], evaluated the of bio-oil and bio-char characteristics obtained from SS using GC/MS. Bio-char obtained from SS pyrolysis had a key role in the elimination of H_2S, NO_x compounds, heavy metal, dye and phenol [25–29]. Additionally, Fonts et al. [13], conducted an in-depth review to study the state-of-the-art of SS pyrolysis for bio-oil production.

However, it is to be mentioned that despite of the interest in the use of MSS for bioenergy production, the associated technologies are still miles away from their final implementation at large scales. Therefore, it is necessary to obtain further information, such as TGA data prediction using machine-learning method and techno-socio-economic assessment (TSEA), concerning MSS pyrolysis in order to find a way for industrial scale-up.

In this regard, comprehensive research regarding the development of MSS pyrolysis plant is scarce, while most of the processing related research has been focused on merely one commercial aspect of the plant, hence the necessity of developing new processes through considering the multi-faceted characteristics of the problem for the implementation of MSS based bio-refineries. Accordingly, the present study was focused on the feedstock characterization, thermo-kinetic studies, developing new machine-learning methodology, evolved gas analysis, and TSEA for MSS pyrolysis and using it as a renewable fuel. Further studied were different experimental, modelling and socio-economic aspects affecting the use of MSS in large scales.

2. Materials and methods

2.1. Sample preparation, proximate, elemental, and HHV analyses

MSS was obtained from holding tank of an urban WWTP site in the west of Tehran (Iran). Samples were sun-dried for 72 h, granulated to a particle size of 3–5 mm, and stored in airtight plastic bags for the subsequent analysis. Later, they were subjected to proximate analyses including volatile matters (VM, %), moisture content (MC, %) and ash (%) which were performed in accordance with standard procedures delineated in ASTM E870-82 (2013) [30]. The fixed carbon (FC, %) content was specified through subtracting VM%, MC% and ash% from 100%. The elemental composition of the sample including Carbon (C), Hydrogen (H), Nitrogen (N), Sulfur (S), and Oxygen (O) was measured using Vario EL III elemental analyzer (Germany), where Argon was used as the carrier gas. Experiments were performed in triplicates and average values were used in calculations. High heating value (HHV) reflects the energy to be released from any biomass upon combustion. Because experimental methods are complex, expensive, and time-consuming, HHV was measured using the linear correlation developed by Nhuchhen and Salam [31]. The results are presented in Section 3.1.

2.2. TGA-DTG experiment for thermo-kinetic studies

Thermal decomposition behavior of MSS was investigated at four different β-values of 5, 10, 30, and 50 °C/min using a TGA instrument (TGA/DSC-1, Mettler Toledo, Switzerland) starting from room temperature (25 °C) to 1000 °C. Nitrogen (99.8% purity) was utilized as the inert environment at a 50 ml/min flow rate in a reaction chamber. In order to reduce or eliminate the mass and heat transfer limitations, a small number of biomass samples (10 ± 0.1 mg) were used in aluminum crucibles (70 μl) in each analysis. The loss of mass, as a function of temperature and its derivative, was recorded via an analytical computer system.

2.2.1. Establishing mathematical model for kinetic analysis

MSS pyrolysis involves a set of complex processes due to the variations in its structural components. Kinetic studies of pyrolysis reaction are important to understanding the thermal decomposition behavior of the biomass and the reaction complexity and mechanism. TGA data is often analyzed using *iso*-conversional model-free based on the Arrhenius equation [10]:

$$\mathrm{k(T)} = A\ \exp\left(-\frac{E_a}{\mathrm{RT}}\right) \tag{1}$$

The transformation rate of biomass from solid-state to volatile-state is described by Eq. (2) [2]:

$$\frac{d\alpha}{dt} = \mathrm{k(T)}f(\alpha) \tag{2}$$

where,

$$\alpha = \frac{m_0 - m_t}{m_0 - m_r} \tag{3}$$

where, α is the conversion degree, m_0 is the initial weight of the sample, m_t is the weight of the sample at time t, and m_r is the residual weight of the sample during the pyrolysis process. Eq. (1) can be inserted into Eq. (2):

$$\frac{d\alpha}{dt} = A\ \exp\left(-\frac{E_a}{\mathrm{RT}}\right)f(\alpha) \tag{4}$$

For non-isothermal condition, Eq. (4) can be written by introducing $\beta = \frac{dT}{dt}$ into the following equation:

$$\beta\frac{d\alpha}{dT} = A\ \exp\left(-\frac{E_a}{\mathrm{RT}}\right)f(\alpha) \tag{5}$$

Now, if Eq. (5) is integrated considering the first reaction order, $f(\alpha) = 1 - \alpha$, and for the initial conditions, $\alpha = 0$, at $T = T_0$, the integral form of the conversion rate equation is generated [11]:

$$\mathrm{G}(\alpha) = \int_0^{\alpha}\frac{d\alpha}{\mathrm{f}(\alpha)} = \frac{A}{\beta}\int_{\mathrm{T_0}}^{\mathrm{T}}\exp\left(-\frac{E_a}{\mathrm{RT}}\right)\mathrm{dT} = \frac{AE_a}{\beta\mathrm{R}}\int_{\mathrm{u}}^{\infty}u^{-2}e^{-u}du = \frac{AE_a}{\beta\mathrm{R}}P(u) \tag{6}$$

Eq. (6) could be solved via different approaches depending on function $P(u)$. In this work, three *iso*-conversional methods (known as model-free methods), namely FWO, KAS, and Starink, were used to specify the E_α and pre-exponential factor (A) as described below.

2.3. Flynn-Wall-Ozawa (FWO) method

FWO method [32,33], an integral *iso*-conversional method, was introduced followed by a few mathematical modifications and Doyle's approximation:

$$\ln(\beta) = \ln\left(\frac{AE_a}{\mathrm{R\ G}(\alpha)}\right) - \frac{E_a}{\mathrm{RT}} \tag{7}$$

Plotting (lnβ, y-axis) versus the inverse of pyrolysis temperature (1/T, x-axis) creates straight lines at different α value with slopes of $-\frac{E_a}{\mathrm{R}}$. Therefore, E_a value is calculated from the slope for the selected values of α.

2.4. Kissinger-Akahira-Sunose (KAS) method

KAS method [34], which offers an improvement in the accuracy of E_α, was applied to Eq. (6); after rearrangement and taking the logarithm of both sides of the Eq. (6):

$$\ln\left(\frac{\beta}{\mathrm{T}^2}\right) = \ln\left(\frac{AR}{\mathrm{E\ \ G}(\alpha)}\right) - \frac{E_a}{\mathrm{RT}} \tag{8}$$

2.5. Starink method

Starink method [35] is based on the following expression:

$$\ln\left(\frac{\beta}{\mathrm{T}^{1.92}}\right) = \ln\left(\frac{AR}{\mathrm{E\ \ G}(\alpha)}\right) - \frac{E_a}{\mathrm{RT}} \tag{9}$$

A graph of $\ln\left(\frac{\beta}{\mathrm{T}^{1.92}}\right)$ is plotted against 1/T where T is the pyrolysis temperature. E_a, is calculated from the gradient of the straight line graph.

2.5.1. Measuring thermodynamic parameters

The A-value (min^{-1}) was calculated using the generalized master–plots method as described in Ref. [36]. The thermodynamic parameters including enthalpy change (ΔH, kJ/mol), Gibbs free energy (ΔG, kJ/mol) and entropy change (ΔS, kJ/mol. K) were calculated using the following equations [37].

$$\Delta H = E_a - \mathrm{RT} \tag{10}$$

$$\Delta G = E_a + \mathrm{RT_m}\ln\left(\frac{\mathrm{K_B T_m}}{hA}\right) \tag{11}$$

$$\Delta S = \frac{\Delta H - \Delta G}{\mathrm{T_m}} \tag{12}$$

where, $\mathrm{K_B}$ is Boltzmann constant (1.381×10^{-23} J/K), h is Plank constant (6.626×10^{-34} Js), $\mathrm{T_m}$ is DTG peak temperature (K), and R is universal gas constant (0.008314 kJ/mol. K).

2.6. SVR model development

Support vector regression (SVR) model was developed using own coding algorithm for a clear understanding of the pyrolytic conditions required for the efficient use of MSS in commercial thermal plants. Fundamentally based on the statistical theory, this technique is employed in computational intelligence associated with the machine-learning methodology. In this model, a sub-optimal discriminant hyper-plane is obtained to maximize the margin between data samples. Normally, a small number of critical boundary samples are selected from each class (in a classification problem), and a linear discriminant function is built [38]. Due to the complexity in the structure of MSS which cause changing the results of TGA curve for different β-value, the input signals in the SVR model related to each β-value were operated independently from each other. Therefore, it means that four models were fitted, one for each β-value. In the present research, SVR model was developed using heating rate (°C/min) and temperature (°C) as input variables, whereas the weight loss (%) at various heating rates against the mentioned temperature points was taken as output data. The experimental data presented in Fig. 1 have been used to fit the SVR model. For data prediction, an algorithm based on radial basis functions (as kernel function) and ϵ-insensitive (as a loss function) was further programmed in MATLAB®R2017b. First, the data set was grouped into a dependent parameter (output/target) and independent parameters

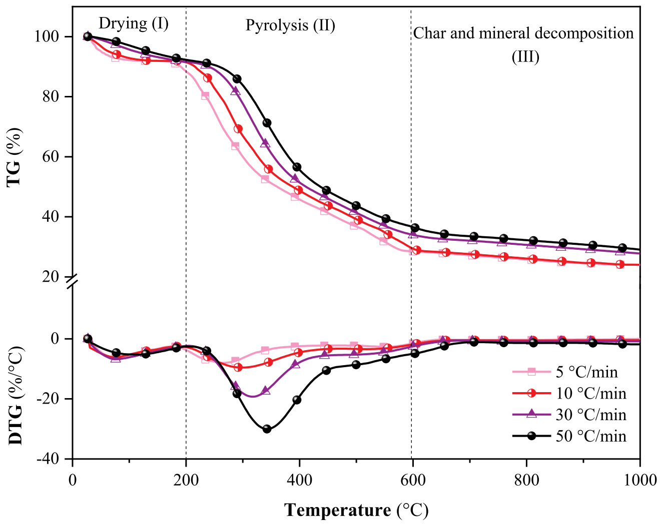

Fig. 1. TGA-DTG curves of municipal sewage sludge (MSS) at different heating rates (5–50 °C/min).

(input), and further normalized to improve the network efficiency. The validation procedure used in the manuscript is the *k*-fold cross-validation where the parameter *k* is variable based on the considered categories. For applying the *k*-fold cross-validation in the defined categories and in order to prevent the leaking and over-fitting phenomenon and increasing the reliability and robustness of the developed model, after shuffling the original data, 20% of the samples were randomly separated as external test data which have not participated in the training process. Then, the remaining of the samples (80%) were entered into the proposed model to find the model's parameters. To better fathom of the model's performance on unseen data, various modes of data division were selected: 99%–1%, 95%–5%, 90%–10%, 80%–20%, 70%–30%, 60%–40%, and 50%–50%. The first number indicated the percentage of training data (e.g. 95% in 95%–5% mode), while the second one showed the percentage of test data (e.g. 5% in 95%–5% mode) which used for the validation of training stage. It should be noted that the model with minimum value of mean square error (MSE) in each of the defined categories has been considered as the best model. The results are presented in Table 2. It would be mentioned that the final evaluation of selected models was conducted using the external test data.

The MSE function, as seen in Eq. (13), was used as an error function to assess the performance of the developed model, while the correlation factor, R^2, as the optimization of the foregoing model, was determined using Eq. (14).

$$\mathrm{MSE} = \frac{1}{n\left[\sum_{i=1}^{n}(\lambda_i - \beta_i)^2\right]} \tag{13}$$

$$\mathrm{R}^2 = 1 - \left[\frac{\sum_i (t_i - o_i)^2}{\sum_i (o_i)^2}\right] \tag{14}$$

where, λ_i : experimental values, β_i :predicted values, n :number of data points, t :target values, and o :output values. Smaller MSE values and higher R^2 values indicate the better performance of the developed model. In order to find out the effect of parameter *k* in the *k*-fold cross-validation procedure and the accuracy of the developed model on unseen data, for the first time, a further analysis was carried out by applying the external test data to the obtained models in all of the considered categories. The results are presented in Table 3.

Table 1
Criteria, key assumptions, and prices for economic analysis.

Parameter	Unit	Value
Reference country	–	France
Feed rate	kg/hour	50
Plant construction period	months	12
Plant economic life	years	20
Discount rate	%	10
Corporation tax rate	%	35
Amortization rate	%	6.7
Inflation rate [39]	%	2
Electricity price [40,41]	€/kWh	0.1
Cooling water price [42]	€/m^3	0.13
Natural gas price [42,43]	€/kWh	0.033
Gross salary for common labor	€/hour	10
CTC for common labor [a]	€/accountable hour	10.4
Effective cost related to common labor [b]	€/effective hour	12.06
Gross salary for skilled labor	€/hour	13
CTC for skilled labor	€/accountable hour	17.5
Effective cost related to skilled labor	€/effective hour	20.3
Effective working time in each shift	hour/year	1575
Number of working shift per day	–	3
Accountable working days [c]	days	261
Paid vacation days	days	25
Public holidays	days	11
Effective working days	days	225
Financing	% owned capital	100

[a] CTC: Cost to company.
[b] Calculated as CTC*261/225.
[c] Saturdays and Sundays were not considered as working days.

2.7. Evolved gas analysis

To investigate the distribution of gasses evolved from pyrolysis process and qualitative characterization of molecular composition, Pyrolysis–Gas Chromatography/Mass Spectroscopy (Py-GC/MS) experiment was performed on a Pyroprobe pyrolyzer (5200 series) connected with GC/MS instrument (6890 GC/5973 MSD, Agilent, USA). First, a 3-mg (±0.1 mg) amount of sample was pyrolyzed at 700 °C, and the pyrolysis products were then transferred to the GC system and separated on the HP-5MS capillary column (30 m × 0.25 mm × 0.25 μm) using Helium as the carrier gas. The temperature of the GC oven was programmed from an initial 70 °C (for 2 min) to 280 °C (for 7 min). MS was carried out under electron ionization of 70eV. The injector temperature and transfer line temperature of MS were 150 °C and 280 °C, respectively. The eluted compounds obtained through GC/MS were determined using the NIST library.

2.8. Techno-socio-economic assessment (TSEA)

TSEA plays a key role in terms of profitable plant design, reflecting the market scenario and social consequences of bioenergy production via pyrolysis process. Table 1 shows the assumptions and parameters used for economic analysis.

The time value of money over the plant lifetime (20 years in this study) was further taken into account. Using cash flow analysis, four economic indicators were considered for economic assessment: net present value (NPV), return on investment (ROI), internal rate of return (IRR), and profitability index (PI).

NPV is the difference between the present value of cash inflows and outflows reflecting the overall returns to the project in today's euros. This difference is generally used for analyzing the profitability of projects, and it can be calculated by Eq. (15) [44]:

$$\mathrm{NPV} = \sum_{t=0}^{r} \frac{F_t}{(1+r)^t} \tag{15}$$

where, F_t (calculated using Eq. (16)) is the net cash inflow during the considered period, F_0 is the initial investment, r is the discount rate (here

Table 2
SVR model performance indices for the TGA data in nitrogen atmosphere excluding the external test data.

Selected categories	R^2 [a]			MSE [b]		
	Train	Test	Total	Train	**Test**	**Total**
99%-1%	1.00000	1.00000	1.00000	2.8553E-04	5.0925E-05	2.8303E-04
95%-5%	1.00000	1.00000	1.00000	1.1553E-04	1.6368E-04	1.9132E-04
90%-10%	0.99998	0.99997	0.99999	3.0753E-04	2.8679E-04	3.0546E-04
80%-20%	0.99996	0.99995	0.99998	3.2008E-04	3.6739E-04	3.2951E-04
70%-30%	0.99993	0.99996	0.99997	2.9003E-04	3.4141E-04	3.0539E-04
60%-40%	0.99992	0.99995	0.99997	4.3887E-04	8.8877E-04	6.1819E-04
50%-50%	0.99991	0.99993	0.99996	3.2139E-04	2.3407E-04	1.3275E-04

[a] $\mathbf{R^2}$, Correlation factor.
[b] **MSE**, Mean square error.

Table 3
SVR model performance indices using external test data.

Model's indices	99%–1%	95%–5%	90%–10%	80%–20%	70%–30%	60%–40%	50%–50%
MSE	5.3126E-04	4.9256E-04	4.6136E-04	3.9521E-04	4.2141E-04	9.3327E-04	6.3413E-04
$\mathbf{R^2}$	0.99995	0.99996	0.99994	0.99998	0.99997	0.99991	0.99992

10%), and t is the considered time period [45].

$$F_t = (1 - t_r)(\text{Sales} - \text{Costs} - \text{Amortization}) - \text{Investment} \tag{16}$$

where, t_r is the tax rate (35% in this study) and amortization rate is 6.7% [46]. In addition, an inflation rate of 2% was considered. A project with positive NPV is regarded as profitable and acceptable, while a project with negative NPV should probably be rejected due to its negative cash flow.

ROI is a decision tool from a business perspective, used for capital budgeting and evaluating the performance of an investment project, estimated by equation (17):

$$ROI = \frac{\text{Net profit}}{\text{Investment}} * 100 \tag{17}$$

On the other hand, IRR is the discount rate corresponding to NPV = 0 as shown in Eq. (18).

$$\sum_{t=0}^{r} \frac{F_t}{(1 + IRR)^t} = 0 \tag{18}$$

As another economic parameter, PI was specified in order to understand the profitability of the designed scenario based on Eq. (19)

$$PI = \frac{NPV}{\text{Initial Investment}} \tag{19}$$

It is clear that a PI = 1.0 is the lowest acceptable value, whereas any value higher than 1.0 indicates that the NPV is more than the initial investment, reflecting project profitability.

In general, projects with higher NPV, ROI, and IRR and PI > 1 are economically more favorable than others [47]. Finally calculated was the payback period, the first year the project starts being profitable (cumulative cash flow is positive). To estimate the capital and production costs, it is necessary to determine the list of the main equipment. The cost of main equipment was identified based on Aspen Capital Cost Estimator (ACCE) or vendor quotes.

The total capital cost was calculated by multiplying the corresponding Lang factors by the main equipment cost as described in Ref. [46]. The total production cost was calculated as the sum of the depreciation including amortization, property tax, and insurance, and the direct production costs such as utilities, labors, transportation, overhead, and so forth. The cost of operating labor, the necessary man-power for the correct process operation, was estimated based on the relative labor rate [46]. The developed plant was assumed to have one-year startup time, 225 effective working days per annum, 20-year plant lifetime, and no equipment salvage value at the end of the plant lifetime. Pyrolysis reactor was designed at 600 °C. The main energy consumption stream associated with the designed plant was calculated using Aspen Plus® (v8.8). The price of pyrolysis products (bio-char, bio-gas, and bio-oil) was obtained from the literature [45].

2.8.1. Alternative scenarios and sensitivity analysis

Three alternative scenarios were designed to assess the economic feasibility performance of the designed pyrolysis plant. In Scenario I, all pyrolysis products, including bio-oil, bio-gas, and bio-char were assumed to be sold in market. In Scenario II, bio-gas and bio-char were combusted to provide energy in pyrolysis reactor, while in Scenario III, bio-gas was combusted to provide part of the required energy in pyrolysis reactor, where the plant was designed without dewatering unit. Fig. 6 depicts the process flow diagram for all alternative scenarios.

A sensitivity analysis was undertaken to identify the key parameters affecting the economic performance of the competing scenario on the project IRR. In order to optimize the key variables, the tornado diagram was selected for a better insight, assuming ±10% variability on each factors once at a time. The investigated parameters were bio-oil selling price, total operation cost, labors cost, feed rate, plant overhead cost, installation cost, main equipment cost, project contingency, bio-gas selling price and bio-char yield for pyrolysis. The results are presented as a tornado plot in Fig. 8.

3. Results and discussion

3.1. Main characteristics of the MSS

The sun-dried MSS was characterized using proximate, elemental and HHV analyses. The VM, ash content, and FC of the feedstock were 52.7 ± 0.3 wt%, 30.9 ± 0.5 wt%, and 6.7 ± 0.2 wt%, respectively. The VM includes compounds released as volatiles during heating, while FC refers to the solid combustible residue left after the sample was completely burned and released of its volatiles. Presence of higher VM in the feedstock indicates the formation of more volatiles able to increase the amount of bio-oil production via pyrolysis process [2,24]. Moreover, the FC content of MSS was lower than the VM content, showing the faster decomposition rate during pyrolysis process and reflecting potential as a highly reactive bio-gas fuel [48]. The main reason for the high ash content of MSS is the influx of silts and sand particles into WWTP causing low HHV [20].

According to elemental analysis, MSS contained 47.4 ± 0.2% C, 6.9 ± 0.1% H, 7.3 ± 0.2% N, 1.2 ± 0.1% S, and 36.3 ± 0.3% O. The high

O content is mainly attributable to highly oxygenated compounds in MSS such as (hemi)cellulose; other organic substances resulted in more easily achieved bio-oil ignition. MSS has high N and S contents as compared to other well-known energy crops [49], which results from the protein fraction of the microorganisms used in WWTP [50]. The C/H and C/O atomic ratios of MSS were 0.57 and 1.74, respectively. The lower value of C/H atomic ratio compared with C/O atomic ratio reflected the presence of a high content of aliphatic hydrocarbon compounds and low amount of polar compounds in MSS, which is in agreement with the tendency of Py-GC/MS analysis as discussed in Section 3.3. The calculated HHV of the MSS, 16.47 ± 0.03 MJ/kg, is comparable with the values of certain lignocellulosic materials (13.5–19.4 MJ/kg) [49], yet higher than those for camel grass (15 MJ/kg) [51], para grass (15.04 MJ/kg) [52] and algal biomass (10.1 MJ/kg) [53]. Moreover, the acceptable range of HHV for the commercial biomasses was expected to be 15–19 MJ/kg [54]. HHV analysis result is in good agreement with the previously reported paper [50]. The high content of VM and C indicates that MSS has a considerable potential for bioenergy production via pyrolysis process.

3.2. Thermo-kinetic studies

3.2.1. Thermal degradation pattern assessed through TGA-DTG curves

Thermal decomposition pattern of MSS was studied via TGA-DTG curves in a temperature range of approximately 25–1000 °C at varying β-values of 5, 10, 30, and 50 °C/min in a N_2 atmosphere as shown in Fig. 1. This figure indicates physicochemical changes taking place during the thermal degradation of MSS into various products (bio-oil, biochar, bio-gas) [55].

TGA curves showed a typical appearance similar to the other biomass such as camel grass [51], perennial grass [56], horse manure [57], sawdust [58], algal biomass [59], textile dyeing sludge and pomelo peel [60], but the temperature ranges were different. Moreover, the slight shift with increase in β showed that β was not able to influence decomposition chemistry. Therefore, a lower β should be utilized to optimize pyrolysis process and obtain pyrolytic product [51].

As shown in Fig. 1, mass loss pattern (TGA curves) of MSS was divided into three distinctive stages. The first stage (I) included temperature of up to 200 °C and all heating rates, indicating the release of retained moisture content within the surface or free bond water and corresponding to endothermic dehydration and vaporization of low boiling components of MSS. During this stage, depending on β, an associated total mass loss of 8.9%–12% was observed. In the second stage (II), with a temperature range of 200–600 °C, referred to as active pyrolysis zone, volatiles such as hydrocarbons, proteins, carboxylic acid, polysaccharides, sugars, and aliphatic compounds were released, and a significant mass loss and thermal transformation occurred (56.6%–61.1%). In this zone, higher molecular weight compounds were fragmented into smaller ones by applying continuous heat [20]. The third stage (III), temperatures above 600 °C, was due to the decomposition of inorganic contents as thermal-stable components at high temperatures such as calcium carbonate mainly associated with the formation of bio-char, which is in line with other studies [22,61].

It is further observed from Fig. 1 that ignition temperature (T_i, 170 °C for 5 °C/min while 230 °C for 50 °C/min) and T_m shifted to higher temperatures with the increase in β, indicating the limitation of mass and heat transfer during the pyrolysis process. Most of the mass loss was observed at temperatures of up to 600 °C, compared to the maximum mass loss achieved at 1000 °C, indicating that no significant mass conversion reactions took place at temperatures higher than 600 °C. Therefore, the suitable temperature range for the thermal conversion of MSS may be from T_i up to 600 °C. Final mass residues ranged from 24.6% to 30.4% at 1000 °C, rendering MSS suitable for bio-char production.

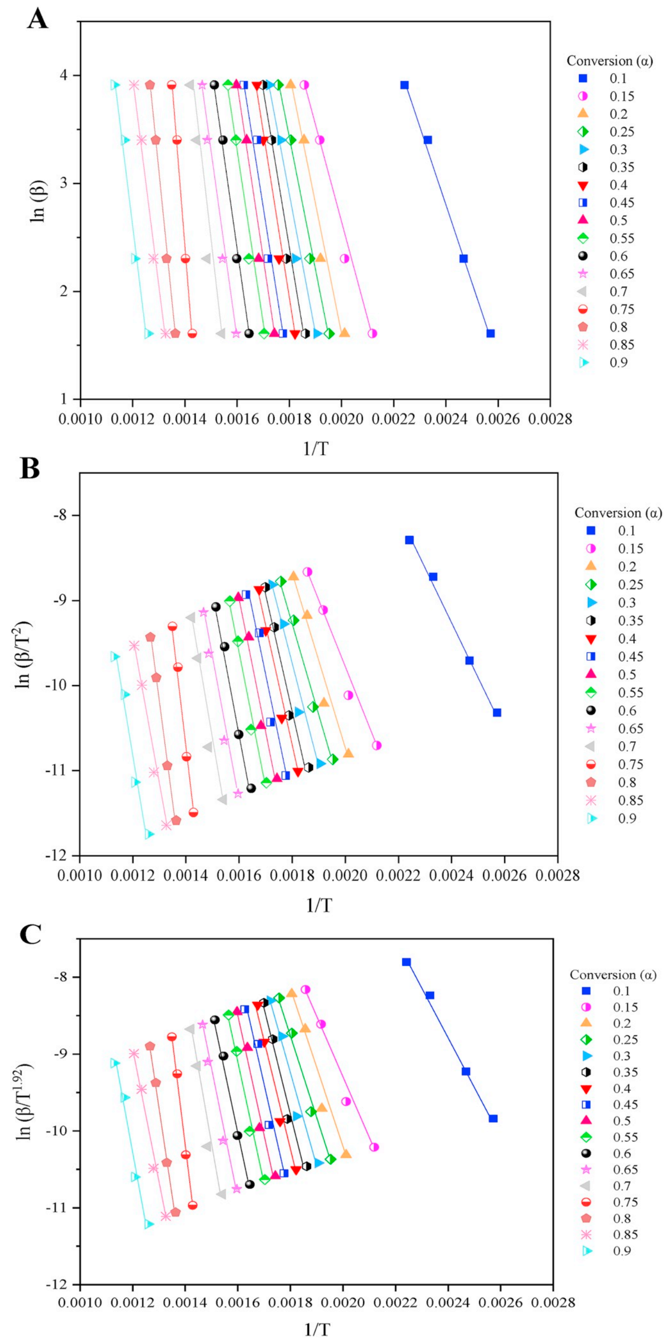

Fig. 2. Arrhenius plots of MSS to calculate activation energy. Where, ln (β), ln (β/T^2), and ln ($\beta/T^{1.92}$) were plotted against inverse of pyrolysis temperature using FWO (A), KAS (B), and Starink (C) methods.

3.2.2. Pyrolysis reaction kinetic and activation energy (E_a, kJ/mol)

The kinetic analysis of the thermal degradation of MSS is an important component in the efficient design of pyrolysis process, which, more often than not, have unknown or convoluted mechanisms [20]. To specify the kinetic and thermodynamic parameters, FWO, KAS, and Starink methods were employed, where, $\ln(\beta)$, $\ln\left(\frac{\beta}{T^2}\right)$, and $\ln\left(\frac{\beta}{T^{1.92}}\right)$ on Y-axis were plotted against the inverse of pyrolysis temperature on X-axis (see Section 2.2.1), respectively, as shown in Fig. 2. E_a, the energy required to break the chemical bonds among molecules in order for the reaction to take place, was calculated by normalizing the TGA data from 0.1 to 0.9 with a step-size of 0.05. The variations of E_a and A versus the α of MSS are shown in Fig. 3, indicating that E_a and A are highly

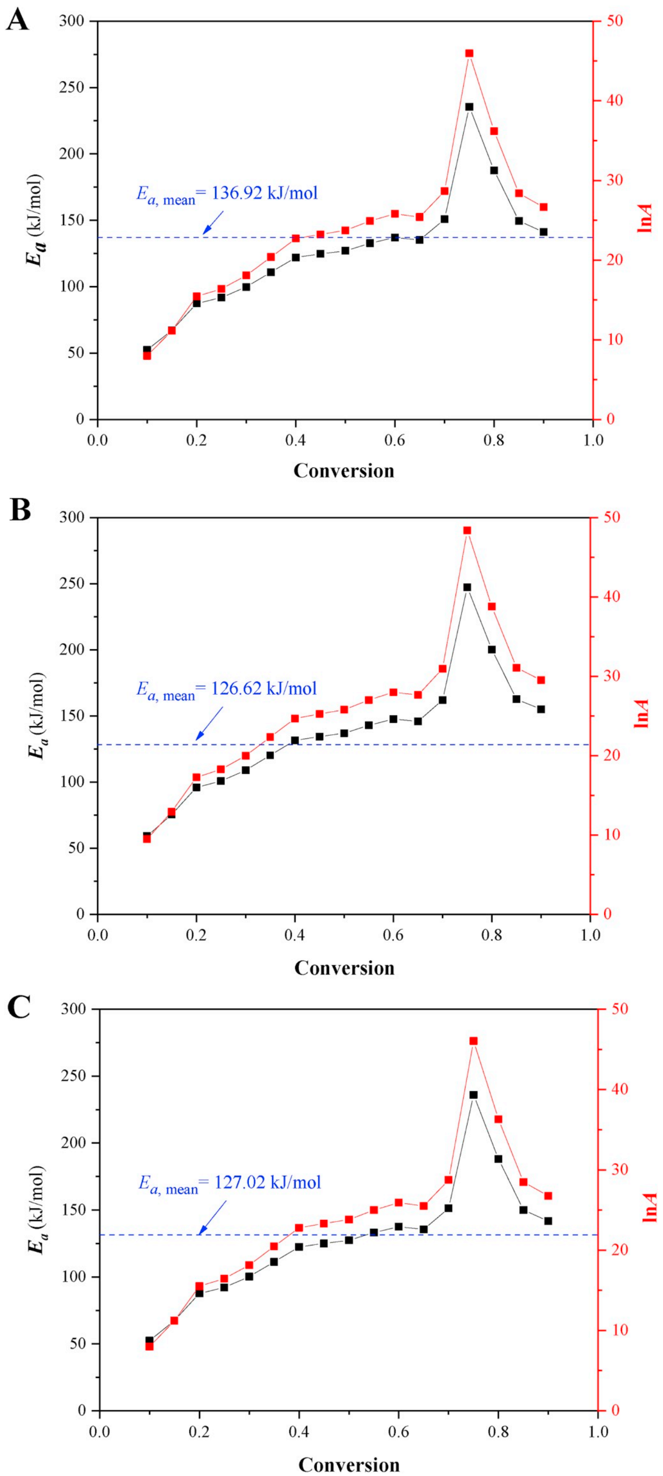

Fig. 3. Activation energy (E_a) and pre-exponential factor (ln$A$) of pyrolysis process of municipal sewage sludge using three different methods FWO (A), KAS (B), Starink (C).

dependent on α values. The changes in E_a at different α values show the existence of a complex reaction mechanism within MSS during de-volatilization [57].

E_a values increased with the increase in α, peaked at α = 0.75 due to the change in the reaction chemistry of inorganic component degradation, and decreased with the further increase in α, which is in line with other study [10]. As shown in supplementary material (SM) Table S1, the E_a values ranged from 59.32 to 247.43 kJ/mol (Avg.

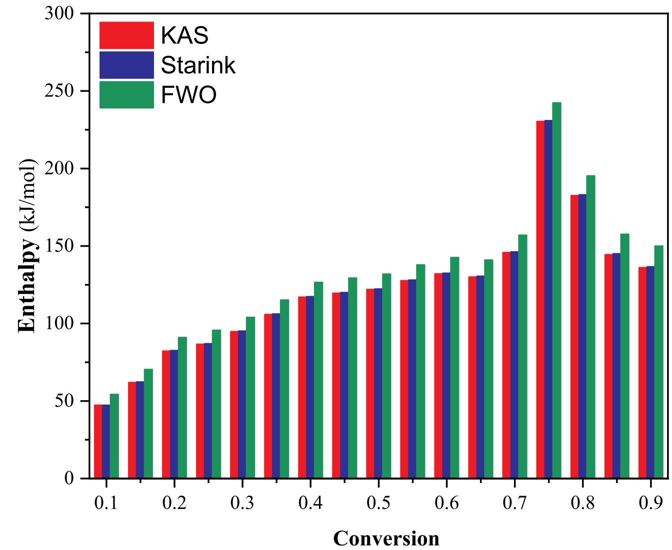

Fig. 4. Enthalpy variations at different conversion points (α = 0.1–0.9) for FWO, KAS and Starink methods.

136.92 kJ/mol), 52.41–235.45 kJ/mol (Avg. 126.62 kJ/mol), and 52.38–235.93 kJ/mol (Avg. 127.02 kJ/mol) determined by FWO, KAS, and Starink methods, respectively. The average values of E_a (126.62–136.92 kJ/mol) were lower than rice husk (221–229 kJ/mol) and elephant grass (218–227 kJ/mol) [62], and cellulose (191 kJ/mol) [36] and higher than cattle manure (122–124 kJ/mol) [63] and red pepper waste (92.93 kJ/mol) [55]. This makes MSS adequate for co-pyrolysis purposes with several other biomass sources under different conditions. Moreover, at α = 0.75, E_a was 235.43–247.43 kJ/mol, indicating that some highly endothermic reactions occurred at this conversion point. The decrease in E_a and A after α = 0.75 shows, the gradual transition of reaction mechanism during the pyrolysis process [64]. $R^2 \geq 0.97$, satisfied the accuracy requirements to calculate kinetic parameters. Fig. S2 (in SM) shows the relationship between α and pyrolysis temperature at different β-values for MSS. In accordance with other studies on different materials, α showed a non-linear relationship with the pyrolysis temperature [52].

3.2.3. *Thermodynamics analysis*

The thermodynamic parameters, including A, ΔH, ΔG, and ΔS, which indicate the pyrolysis behavior of the feedstock, were calculated at a heating rate of 5 °C/min.

3.2.3.1. Pre-exponential factor (A, min^{-1}). The A-value is the number of collision per time unit, obtaining the proper orientation for reaction to take place [57]. A-values elucidate the reaction chemistry, which is critically important when optimizing the pyrolysis process; $A<10^9$ min^{-1}, on the other hand, mainly shows the surface reaction, and $A\geq10^9$ min^{-1} indicates a complex reaction when the reaction is not dependent on the surface area [65]. As shown in Table S1 of SM, the wide range of A-values indicates the complicated chemical reaction during pyrolysis due to the complex structure of MSS, previously observed for the pyrolysis of SS [20], red pepper waste [55], and *Helianthus tuberosus* [10]. At α = 0.75, A-value of 10^{19}–10^{21} was obtained owing to the extra energy required during the degradation of inorganic component. Similarly, a higher E_a was observed at α = 0.75, which requires more heat to be transferred for a higher molecular collision. It is worth mentioning that MSS has a complex composition that requires different energy levels at different conversion points. Fig. S1 (SM) shows the slopes of the straight lines were used to determine the corresponding E_a and A-value at each conversion point using selected kinetic methods.

3.2.3.2. Enthalpy change (ΔH, kJ/mol). Enthalpy change, Δ*H*, reflects the amount of energy consumed by the feedstock during pyrolysis process for its conversion into various products [66]. The overall positive value for Δ*H* illustrates that the pyrolysis of MSS is an endothermic reaction, where heat is absorbed by the bonds to break and form a new one. Average Δ*H* values of MSS pyrolysis were 131.90 kJ/mol, 121.60 kJ/mol, and 121.99 kJ/mol, obtained by FWO, KAS, and Starink methods, respectively. The difference between E_a values and Δ*H* elucidates the likelihood of occurrence of pyrolysis reaction [34]. A meagre difference of ~5 kJ/mol was observed between E_a and Δ*H* values (Table S1), showing that there existed a small potential energy barrier to achieve the product formation, which is in accordance with other results [34]. Moreover, as shown in Fig. 4, the Δ*H* values were consistent with one another at all conversion points, reflecting the reliability of the data.

3.2.3.3. Gibbs free energy (ΔG, kJ/mol). Gibbs free energy, Δ*G*, exhibits the amount of available energy from feedstock upon pyrolysis. The Δ*G* values varied from 155.96 to 163.13 kJ/mol, 156.21–163.75 kJ/mol, and 156.20–163.75 kJ/mol as determined by FWO, KAS, and Starink methods, respectively, showing the considerable bioenergy potential of the selected biomass through pyrolysis. Moreover, the positive value for Δ*G* (Table S1) reflects, the energy produced from the biomass pyrolysis. The average of available energy (159.19–159.61 kJ/mol) was higher than the red pepper waste (146.82 kJ/mol) [55] and horse manure (153.25 kJ/mol) [57], but lower than camel grass (174.49 kJ/mol) [51] at the same conversional points.

3.2.3.4. Entropy change (ΔS, kJ/mol.K). Entropy change, Δ*S*, a direct reflection of the disorder degree of a system, elucidates the closeness of a system to its thermodynamic equilibrium in response to different conditions. As shown in Table S1 of SM, Δ*S* values were more negative (as low as −0.193 kJ/mol· K) and less positive (as high as 0.143 kJ/mol· K); more negative values indicate that the disorder degree of products is much lower (more organized) compared to the initial reaction state, while positive values show otherwise. Positive Δ*G* and negative Δ*S* indicated that the thermal conversion of MSS occurred in a non-spontaneous process, where the occurrence was endergonic. Δ*S* value at the conversion point of 0.75 is approximately 0.123–0.143 kJ/mol. K, reducing the formation of organized products such as bio-char. Moreover, a wide range of Δ*S* values indicates a complex reaction chemistry.

3.2.4. Validation of pyrolytic behavior of MSS using SVR model

To further fathom the pyrolysis of the selected biomass, SVR model, a renowned regression algorithm, was developed to predict and validate the accuracy of the TGA data at four different heating rates. In order to prevent the over-fitting and leaking phenomenon, 1%–50% of data (as described in Section 2.3) were separated as test data after shuffling the original data. The remaining data (99%–50%) were used for training stage in the proposed modeling process, where the separated external test data would only be employed in the final evaluation of model, not the training stage, ensuring that the evaluation of the system is not based on the data utilized by the developed model in the training process. By so doing, leaking and over-fitting of the testing-set data to the procedure for training stage was prevented.

First, the whole data set was grouped into a dependent parameter (output/target) and independent parameters (input) for MSS. The data set was further normalized and randomly divided into training data and test data after separating the external test data. As shown in Fig. S3 (SM), regression plot of training, test and total data steps along with error histogram were constructed using SVR model output. The predicted variable was weight loss (%) at various heating rates against the mentioned temperature points. Therefore, in each category of data division, the values predicted using the developed SVR model and the real values were used to plot the correlation figures (see Fig. S3 in SM).

Even by reducing the number of training data, the R^2 for the SVR model is close to unity (Table 2), signifying a good model fit of the predicted and experimental values. Additionally, the error histogram appears to be normally distributed for a major part of the experimental dataset points. As presented in Table 2, SVR model for all categories, carried out sufficient iterations with optimal performance and minimum possible MSE. Based on the results, the SVR model is a good feature to understand and envisage the pyrolytic behavior of MSS required for the efficient design in commercial pyrolysis plants. As shown in Table 3, the model obtained from the category of 80%–20% ($k = 5$) shows the best result due to its lower MSE and higher R^2. Therefore, this category of data division has been applied for training of final model used for the TSEA.

3.3. Evolved gas analysis

The Py-GC/MS analysis was carried out to assess the volatiles evolved from the pyrolysis of MSS. The main components identified upon MSS pyrolysis, as well as its chemical formula, and Chemical Abstracts Service (CAS) number are listed in Table S2 in SM. Pyrolysis of MSS, as a complex mixture, generated compounds from a wide variety of molecular weights (from methylamine, *N*,*N*-dimethyl, $M_w = 59$ to *n*-C_{33}, $M_w = 457$), containing a large number of aromatic and oxygenated compounds. All identified compounds can be grouped into the following category: aromatic hydrocarbon (such as toluene, benzene, styrene, phenol, and their alkyl derivatives); alkanes and alkenes (aliphatic compounds) with carbons ranging from C_9 to C_{33}; furans (such as 2,5-dimethyl-furan, 2-furanmethanol); nitrogen-containing compounds (such as pyridine, 4-methylbenzonitrile, 7-methyl-Indole, 1-octadecanamine); alcohol (such as 1-tetradecanol); sulfur compounds (such as sulfone-methyl phenyl, acetamide, *n*-sulfanilyl); steroid (such as cholestene); Polycyclic hydrocarbon (such as pyrene, naphthalene, 1-methylnaphthalene, 2-fluorenecarboxaldehyd, benzo(a)pyrene 4,5-oxide), and esters (such as tridecanedioic acid, dimethyl ester). According to the Maillard reaction, pyridine, 4-methylbenzonitrile, 1-octadecanamine, and other *N*-containing compounds were derived from the reactions of the carbonyl and amino functional groups of protein fraction of the microorganisms used in WWTPs [67]. The generation of benzene (and its derivatives) and toluene was attributed mainly to the decomposition of (hemi)celluloses and the protein structure of MSS. Different alkane compounds such as *n*-nonane and *n*-decane and were derived from the degradation of organic matter in MSS [68]. Aromatic hydrocarbons were generated from the secondary cracking reactions of proteins and alkene (such as 1-undecene) through an intermolecular reaction [24]. The HHV of toluene, benzene, ethylbenzene, and *n*-nonane were 40.6, 41.8, 40.8 and 44.3 MJ/kg comparable with gasoline and diesel with 47.3 and 44.8 MJ/kg, respectively, confirming that MSS has the efficient energy-yielding capacity in terms of usage as a promising feedstock to produce bioenergy via pyrolysis in a cost-effective and environmentally-friendly manner.

3.4. Techno-socio-economic assessment (TSEA)

TSEA was carried out to provide a realistic analysis regarding different aspects of MSS pyrolysis plant which depends on the biomass characteristics, plant size, cost of the required energy, production capacity, and regional implications. For this purpose, three alternative scenarios were designed. In Scenario I, all of pyrolysis products were assumed to be sold in market, whereas in Scenario II, bio-gas and bio-char were combusted in pyrolysis reactor. In Scenario III, bio-gas was combusted in pyrolysis reactor where the plant was designed without dewatering unit.

The results pertaining to the comparison of the alternative scenarios over the 20-year planning horizon were discussed by key economic parameters. The feed rate of MSS was 50 kg/hr with an effective operating time of 4725 h/year for all alternative scenarios. Fig. S4 in SM depicts the cash flow diagram after taxes over the plant lifetime for

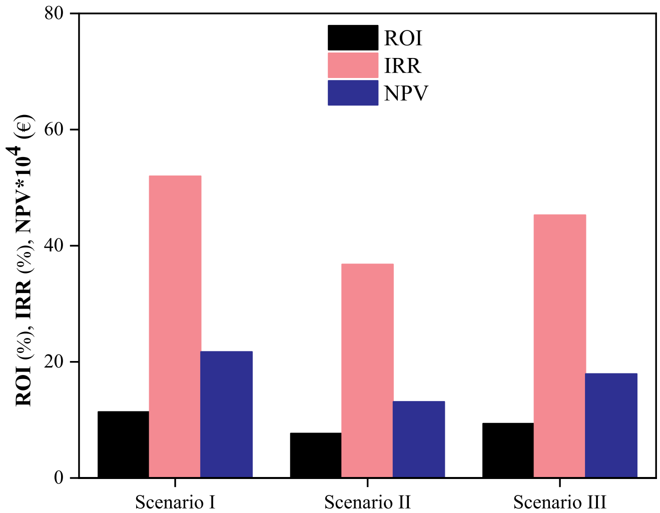

Fig. 5. Return on investment (ROI), Internal rate of return (IRR), and Net present value (NPV) of pyrolysis plant for alternative scenarios.

Scenarios II and III. As shown in Fig. S4, the annual income was reduced by € 5318 and € 2870 while the total production costs were increased by 3.5% and 1.36% due to increase in energy cost for the Scenarios II and III in comparison to Scenario I, respectively. As illustrated in Fig. 5, the economic parameters associated with Scenario I were higher than other scenarios, showing a better profitability in this scenario. In Scenario I, the related project IRR was increased by 41.3% and 14.8% and the ROI was increased 48%, 21.3% in comparison to Scenarios II and III, respectively. Thus, Scenario I reached the highest profitability among the three investigated scenarios in terms of IRR, ROI, and NPV, hence considered as an optimistic and more competitive scenario in this study.

As illustrated in Fig. 6A, the pyrolysis plant of Scenario I consists of six main sections: dewatering, drying, pyrolysis reactor, cyclone, condenser, and storage tank. The dewatering unit is required for decreasing the liquidity of the feedstock up to ~20%, containing 6% of the main equipment costs. Drying unit is used to reduce the moisture of the MSS to less than 10% preparing it for the pyrolysis process. The required electricity power was 5.95 kW (21.43 MJ/h) for the operation of the designed plant based on Scenario I (Table 4). The current model required cyclone to remove any residuals following pyrolysis, totaling 26% of the main equipment cost. Storage tank is employed for the storage of bio-oil which is made of stainless steel costing € 5176. Water was assumed to be accessible with no permit fees and taxes in any alternative scenario. Combustion of bio-gas for energy input in pyrolysis plant (Fig. 6B and C) contributes to the increase in the production costs, resulting in lower IRR and ROI values calculated for Scenarios II and III.

Tables 4 and 5 depict the costs (including capital and production costs) for starting up the pyrolysis plant and its associated annual revenue for the optimistic scenario, respectively. Installation and production costs of the pyrolysis plant were calculated based on data in Ref. [46], which were 14.2% and 80.4% of the total costs. They were used to evaluate the economic viability of the designed plant. Operating labor cost and plant overhead cost comprised 82.4% of the total production costs, and the operating labor cost was the largest contributing factor in terms of operational costs. Dominating factors within the installation costs included project contingency and construction expense, consisting 10.4% and 9.5% of total capital cost respectively.

Given that unlike other feedstock, MSS is produced as a waste in high

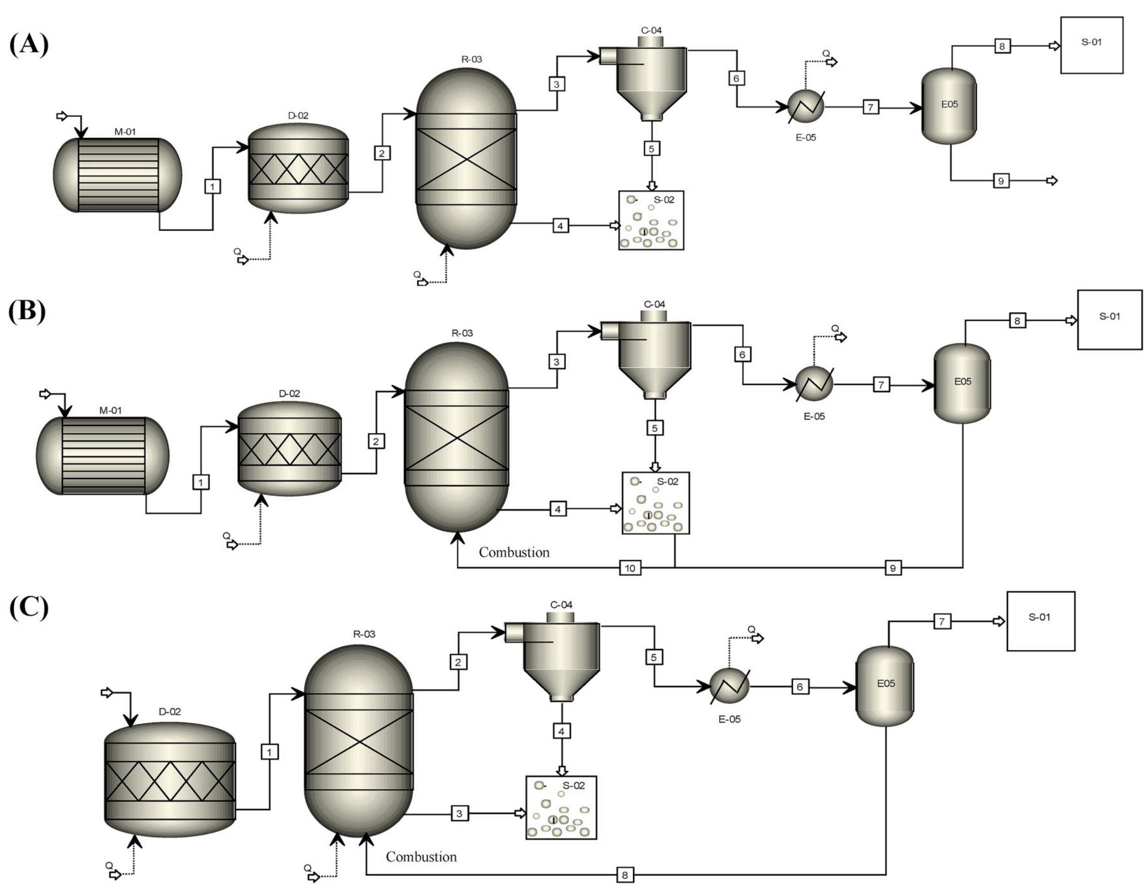

Fig. 6. Process flow diagram for different scenarios considered. (A) Scenario I, (B) Scenario II, and (C) Scenario III. (M-01: Dewatering; D-02: Dryer; R-03: Pyrolysis reactor; C-04: Cyclone; E−05: Condenser; S-01: Bio-oil storage tank, S-02: Bio-char gathering).

Table 4
Capital and production costs for pyrolysis plant (Scenario I).

Components	Costs (€)//(€/year)	Description
Capital Costs		
Main equipment (E)		
Dewatering	1126	To decrease liquidity up to ~20%
Dryer	1441	To reduce the moisture <10%
Storage tank	5176	To storage bio-oil and made of stainless steel material
Pyrolysis reactor	4501	To pyrolysis the feedstock
Cyclone	5042	To remove any residual after pyrolysis process
Condenser	2101	To obtain bio-oil
Total	**19,387**	
Installation (I) [a]		
Direct		
Buildings (including services)	5622	29% E
Instrumentation and controls	5041	26% E
Electrical systems	1939	10% E
Piping	6010	31% E
Insulation	969	5% E
Indirect		
Engineering and supervision	6204	32% E
Construction expense	6592	34% E
Licensing	3877	20% E
Legal expense	775	4% E
Contractor's Fee	3684	19% E
Project contingency	7173	Due to inherent uncertainties in scale-up (37% E)
Auxiliary services	1939	10% E
Total	**49,825**	
Production Costs [a]		
Feedstock transportation[b]	14,354	To transport feedstock towards pyrolysis plant
Utilities[c]	4218	Include natural gas, electricity, and cooling water
Operating labor (O.L)[d]	152,901	
Maintenance and repairs (M&R)	9129	Linear amortization 6.7%, and inflation rate 2%
Operating supplies	4149	21.4% E
Laboratory charges	15,290	For product quality control (10% O.L)
Local property taxes	775	4% E
Property insurance	1939	1% E
Plant overhead costs[e]	81,015	
Total	**283,770**	

[a] Calculated according to the percentage method as described in Ref. [46].
[b] Calculated based on data in Ref. [71].
[c] Utilities cost calculated as follow: Electricity cost: 5.95 kW*0.1 €/kWh*4725 h/year (Electricity is needed as follows: dewatering unit (0.89 MJ/h), dryer (1.78 MJ/h), pyrolysis reactor (34 MJ/h), cyclone (0.56 MJ/h), condenser (−15.8 MJ/h).)- Cooling water cost: 5187 m^3/year*0.13 €/m^3 (The amount of cooling water for the process was calculated using Aspen Plus® (v8.8))- Natural gas cost: 4.7 kW*0.033 €/kWh*4725 h/year (The amount of natural gas for the process was calculated using Aspen Plus® (v8.8)).
[d] Labour costs were calculated as 12.06 €/hr*4725 h/year and 20.3 €/hr*4725 h/year for common and skilled labours, respectively, considering the effective value for working time and cost.
[e] Including general plant overhead, payroll overhead, packaging, safety and protection, recreation (50% (O. L + M&R)).

amounts (as described in the Introduction), biomass production cost (including growth and harvest), as the most influential parameter in microalgae biofuel production system [69,70], was not considered in this study. Cost of transporting MSS from WWTPs to the pyrolysis plant site was, determined based on literature [71]. However, the NPV was positive for all alternative scenarios, and more positive in Scenario I compared with others, showing that this scenario results in the most optimal pyrolysis plant. NPV of the pyrolysis plant was € 217,305 in

Table 5
Annual sales of pyrolysis products (Scenario I).

End products	Production capacity (tone/year)	Selling price [a] (€/kg)	Sell (€/year)
Bio-char	66.150	0.037	2448
Bio-gas	63.787	0.045	2870
Bio-oil	106.312	3.130	332,757
Total			**338,075**

[a] Price of end products (bio-char, bio-gas, and bio-oil) were obtained from Ref. [45].

Scenario I over 20 years, implying that the project is feasible at the designed capacity for the plant. The payback period for all alternative scenarios was nearly 3 years, which shows the financial viability of the designed plants. ROI of the competing scenario was 11.4%, which is higher than the discount rate (10%), indicating the profitability of the project.

PI values were 3.14, 1.90, and 2.59 for the Scenarios I, II, and III respectively, showing that all alternative scenarios are economically sustainable (PI > 1). Moreover, the PI value of Scenario I was higher than others, hence the most profitable. In all alternative scenarios, the project was feasible provided all economic indicators were adequate, meaning IRR>0, ROI>0, NPV>0, PI > 1, and payback time<20 (a 20-year plant lifetime was hypothesized). Fig. 7 shows the cash flow after taxes for Scenario I over the plant lifetime (20-years) where the costs, sales, and cumulative cash in the 20th year were € 427,172, € 482,854, and € 636,334, respectively.

Economic analysis showed that it is economically feasible to produce bioenergy from MSS pyrolysis, which is in line with the literature [72]. In addition, it was shown that microalgae pyrolysis plant (for feed rate of 200 tons/year) is not economically feasible due to its negative NPV, while the production cost of microalgae has to be reduced [40,45].

The most important social benefits associated with MSS pyrolysis plant are as follow: (direct and indirect) increase in employment, food security by avoiding competing to competition with the food industry, rural development by providing new income opportunities, health improvement as a result of reduction in the use of fossil fuel, support of related industries, regional growth, and export potential.

3.5. Sensitivity analysis

Sensitivity analysis was conducted to identify major operational and capital factors impacting the economics of the project, measured by increasing the input factors by ±10%. However, all other input factors were kept constant in order to detect the effect of a selected factor on project IRR. Fig. 8 shows the result of sensitivity analysis evaluating the impact of key variables contributing to project IRR in Scenario I.

Changes in the selected factor associated with a 10% increase in input value are indicated on each side of the baseline using blue color, while 10% decrease in input variable is indicated with red color. The results show that variations in the bio-oil selling price had a higher impact than a similar change in other key economic factors. The increase in the price of bio-oil increased the IRR to 83.4%, while the decrease in the input value decreased the IRR to 17.4%. The decrease in the total production cost from the baseline 283,770 €/year to 255,393 €/year increased the project IRR to 79% and substantially increased the project profitability. However, increasing this factor to 312,147 €/year reduced the project IRR to 22.9%. By optimizing these two factors, small changes can dramatically alter the results of the economic performance of the project. Labors cost and feed rate were the next two most significant factors influencing project IRR. By increasing the feed rate to 55 kg/hr, IRR of the project increased from a baseline of 52%–66.7%. Contrarily, decreasing the feed rate to 45 kg/hr lowered the IRR of the project to 44.6%. Reducing the costs associated with plant overhead cost and installation cost helped increase the IRR of the project by 13.8% and

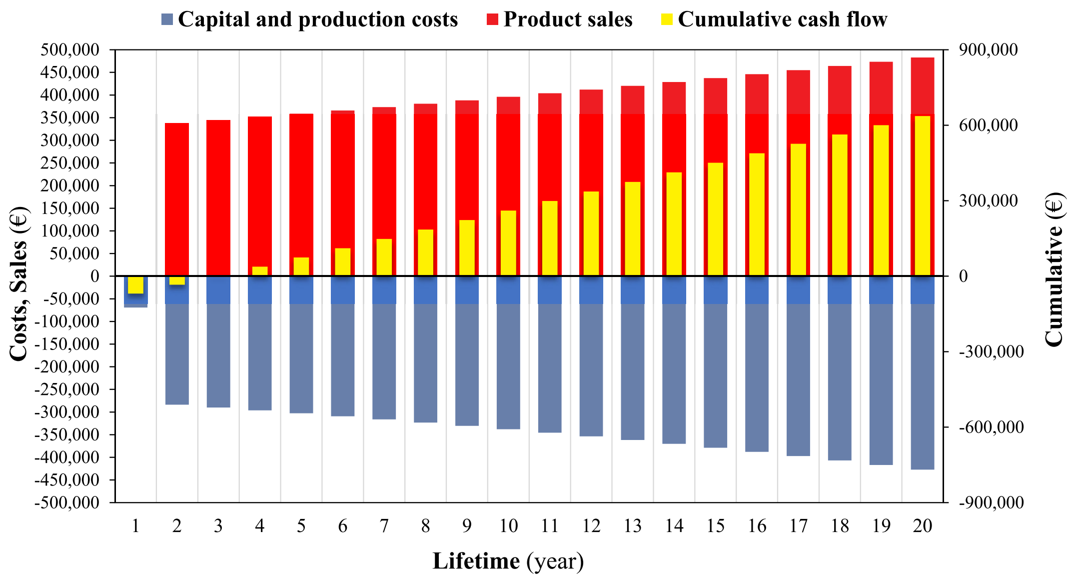

Fig. 7. Annual cash flow diagram of Scenario I over the project lifetime.

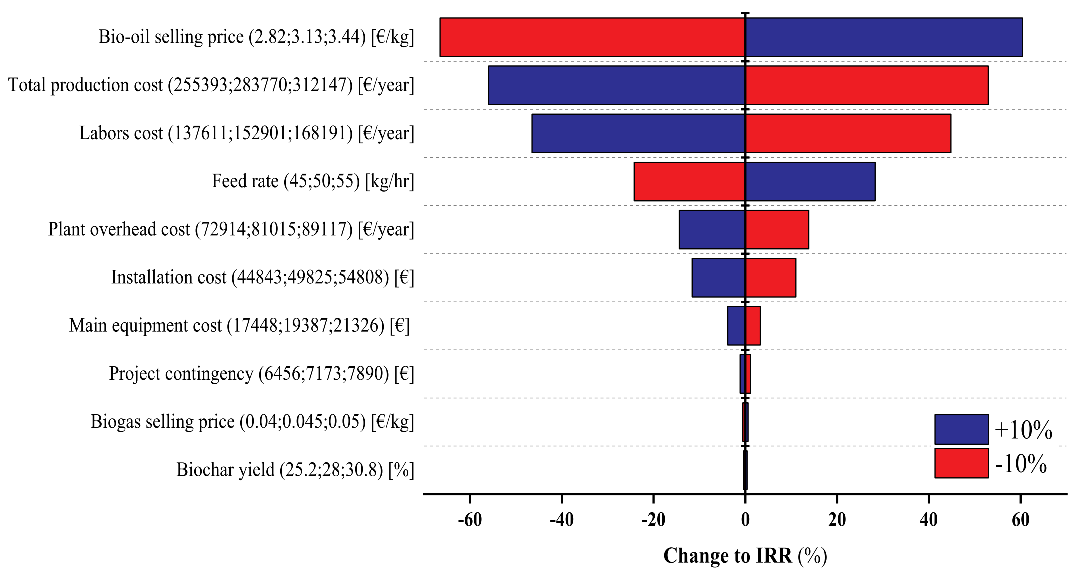

Fig. 8. Sensitivity analysis for important variables in Scenario I of MSS pyrolysis plant using ±10% variability.

11%, respectively; however, they were not as significant as the results obtained with the bio-oil selling price. It was clearly observed that plant overhead cost comprised nearly all of the installation cost. Moreover, increasing the main equipment costs by 10% resulted in a IRR decrease of 3.85%. The remaining factors including project contingency, bio-gas selling price, and bio-char yield had a relatively small impact on the IRR of the project.

3.6. Future perspective

The rapid increase in MSS generation, due to population increase, has rendered it a unique source of renewable energy with a significant potential to produce different products via pyrolysis. MSS pyrolysis is generally regarded as an economical approach to producing bioenergy among other thermochemical platforms for the following reasons: zero-waste process, marketable products, lower GHGs emission, green industry, and a high amount of available feedstock [73]. Despite these strengths, there exist few commercial plants converting MSS into value-added products. To extend the application of this route and make it cost-effective, the following aspects are to be considered:

- Further processing to utilize bio-gas instead of common fuel in a different industry
- Considering the potential application of bio-char for the removal of toxic compounds in industry, remediation of pollution in soil, air and water, and use as fertilizer [74].
- Upgrading the bio-oil into transportation fuel
- Improving pyrolysis reactor configurations
- Appropriately locating pyrolysis plant sites to reduce the transportation cost

Therefore, prior to the application of derived bio-oil, research and innovation are required for upgrading. More studies have to be conducted to investigate the potential TSEA impacts of this waste on bioenergy production. Due to the finite reserves of fossil fuels and their environmental problems, MSS application in bioenergy production should move from small-scale toward large-scale.

4. Conclusions

Pyrolysis of MSS as a free, non-toxic and accessible biomass, was conducted using TGA and Py-GC/MS analysis so as to find its bioenergy potential. The pyrolytic behavior of this waste followed by three devolatilized stages, identified from TGA-DTG curves, indicated the thermal stability of MSS. The first stage showed the release of retained moisture, the second depicted major biomass loss due to the release of volatile organics, and the third showed mineral degradation mainly associated with the formation of bio-char. Thermo-kinetic study was done through FWO, KAS, and Starink, methods, shown to be consistent with one another and applicable to simulating the industrial scale of the pyrolysis plant of MSS. E_a, ΔG, and HHV values showed the remarkable bioenergy potential of the low-cost biomass. Positive ΔG and negative ΔS showed that the thermal decomposition of MSS is a non-spontaneous process. The results obtained from the SVR model showed that the model is capable of directly predicting the thermal characteristics of MSS. Best-fit plots were obtained from the developed model by comparing the predicted and real data. The pyrolysis gas analysis was conducted using Py-GC/MS, indicating the production of valuable chemicals including aromatic and aliphatic hydrocarbons. Moreover, the results of TSEA confirmed the feasibility and profitability of MSS pyrolysis plant at a large-scale in both economic and social dimensions for all alternative scenarios. Moreover, Scenario I was selected as an optimistic scenario due to its higher IRR, ROI, and NPV, hence the highest profitability among other scenarios.

Author contribution

Hossein Shahbeig: Conceptualization, Methodology, Validation, Software, Investigation, Visualization, Formal analysis, Resources, Writing-original draft, Writing-review &editing. Mohsen Nosrati: Conceptualization, Data curation, Methodology, Writing-review &editing, Supervision, Project Administration.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.rser.2019.109567.

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